Little Lab Coats

☀️ The Sun — Our Closest Star

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 1 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Earth Science — Sun, Moon & Stars 🏠 **Materials:** Household only

⚠️ Sun Safety — Read Before Starting!

Never look directly at the sun — not even for a split second. The sun's light is so powerful it can permanently damage your eyes. This lesson uses *shadows* as a safe way to study the sun — we never look at the sun itself. Please review this rule with your child before going outside.

- Always keep your back to the sun when doing shadow observations
- If you feel the urge to look up — look at the ground instead
- Wear sunscreen and a hat on sunny days

The sun, Earth, and its shadow — a system that creates our days and nights

📋 **NGSS Standard: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Describe* the observable pattern of the sun's daily motion across the sky
- *Explain* why the sun appears to move from east to west each day
- *Connect* shadow length and direction to the sun's position in the sky

🔬 The Big Science Idea

The sun appears to move across the sky each day — rising in the east every morning and setting in the west every evening. But here's the surprising truth: *the sun isn't moving!* It's *Earth* that's spinning.

Earth rotates (spins on its axis) once every 24 hours. As we spin, the sun appears to travel across the sky — just like when you spin around in a circle and the trees around you seem to move even though they're standing still. We call this apparent motion.

Because of the sun's position in the sky, shadows change throughout the day. In the morning and afternoon, shadows are long. At midday (when the sun is highest), shadows are shortest.

> 💡 **Key vocabulary:** sun, rotate, orbit, shadow, east, west, midday, apparent motion

📦 Materials

🏠 What You'll Need

- A sunny day (or a sunny window)
- A stick, pencil, or small object to cast a shadow (a "gnomon")
- A large piece of white paper or poster board
- Tape (to secure paper outside if needed)
- A marker or pencil to trace shadows
- A ruler or measuring tape
- A watch or clock
- This worksheet (printed)
- Crayons

No sunny day? Use a lamp at a fixed position indoors as a "sun" and a small object to cast shadows at different lamp angles to simulate morning, midday, and afternoon.

🔬 Let's Investigate — Shadow Tracking!

Step 1 — Set Up (Morning, before 10am)
Place your paper on a flat, sunny spot outdoors (or by a sunny window). Stand a stick or pencil upright in the center — hold it straight or tape it if needed. *Without looking at the sun*, note that the sun is low in the sky. Trace the shadow on the paper and write the time. Measure the shadow's length and note which direction it points.

Step 2 — Midday Observation (Around noon)
Return to the same spot at midday. Trace the shadow again using a different color. Measure its length. Notice: is it longer or shorter than the morning shadow? Which direction does it point now?

Step 3 — Afternoon Observation (2–4pm)
Return once more in the afternoon. Trace the shadow a third time in yet another color. Measure and compare. The shadow should now point in a different direction from the morning shadow.

Step 4 — Record & Discuss
Fill in the Shadow Tracking Record (on the Student Worksheet page). Look at all three shadow tracings together and answer: What changed? What pattern do you notice?

💡 **What to expect:** Morning shadows point roughly west (sun is in the east). Midday shadows are shortest and point roughly north (sun is highest, due south). Afternoon shadows point roughly east (sun is in the west). The shadow sweeps in a wide arc — this is Earth rotating, not the sun moving!

💬 Discussion Questions

🗣️ **Talk about these together:**

- What happened to the shadow's length from morning to midday to afternoon?
- What happened to the direction the shadow pointed?
- If a shadow is very short, where is the sun in the sky?
- If a shadow is very long, where is the sun?

🌟 **Big Question: "The sun looks like it moves across the sky. Is it actually moving?"**

No! The sun stays relatively fixed while Earth spins. Earth rotates once every 24 hours, so from our perspective on the spinning Earth, the sun appears to move from east to west. It's like watching trees go by from a moving car — the trees aren't moving, you are. Ancient people thought the sun moved around Earth (the geocentric model), but we now know Earth orbits the sun (the heliocentric model).

🚀 Extensions

- 🕐 **Shadow Clock:** Try using your shadow tracings to tell time! Ancient people built sundials using this exact idea.
- 🗺️ **Compass Connection:** If you have a compass, check which direction the midday shadow points. It should point roughly north (in the Northern Hemisphere) — this is how explorers used the sun for navigation.
- 📅 **Seasonal Compare:** Do this activity again in 3 months. Are the shadows the same? (Hint: they won't be — connect to Lesson 4!)
- 📚 **Read Aloud:** *"The Sun: Our Nearest Star"* by Franklyn Branley

👩‍🏫 Parent/Teacher Notes

- **Safety first:** Reinforce the no-looking-at-the-sun rule before and during the activity. Make it a habit, not a warning.
- **Timing flexibility:** You don't have to do all three observations in one day. You can do morning one day and midday and afternoon another — just keep the paper setup in exactly the same spot.
- **Common misconception:** Many children (and adults!) believe the sun moves around Earth. This is a great chance to introduce heliocentric thinking without overwhelming — the key idea is just "Earth spins, so the sun seems to move."
- **Grade 1 level:** Don't worry about precise compass directions. The pattern to internalize is: shadow moves throughout the day, shortest at midday.
- **If no sun:** The lamp simulation works well. Hold the lamp at a low angle (morning), high angle (midday), and low angle from the other side (afternoon). Let the child hold the object and trace the shadows.

☀️ Student Worksheet — Shadow Tracker

Unit 1, Lesson 1 — The Sun

Name: ________________________________ Date: ________________

📏 Shadow Tracking Record

Fill in the table as you make each observation. Use a different color for each time of day!

Time of Day	Approximate Time	Shadow Length	Direction Shadow Points	Color Used
Morning				🔴 Red
Midday				🔵 Blue
Afternoon				🟢 Green

🎨 Draw Your Shadows

Draw a bird's-eye view (looking down from above) of your shadow tracings. Show all three shadows coming from the same center point. Label each one!

Draw your morning, midday, and afternoon shadows here

Bird's-eye view of shadow tracings — label morning 🔴, midday 🔵, afternoon 🟢

✏️ Answer These Questions

1. Which shadow was the *shortest*? ______________________ Why do you think so? ______________________
2. Did the shadow move? (circle) **YES** / **NO** — What made it move? ______________________
3. True or false: The sun moves across the sky. (circle) **TRUE** / **FALSE**

4. What is *actually* moving? ________________________

⭐ Think Like a Scientist

A scientist makes a *prediction* before an experiment. What do you predict will happen to the shadows if you do this activity again *at night* with a flashlight pretending to be the sun?

✂️ PARENT ANSWER KEY — Detach before giving worksheet to students

📏 Shadow Tracking — Expected Results

- **Morning:** Shadow is long, points roughly west (away from the rising sun in the east)
- **Midday:** Shadow is shortest, points roughly north (sun is highest in the southern sky)
- **Afternoon:** Shadow is long again, points roughly east (sun is sinking in the west)

✏️ Worksheet Answers

1. The *midday* shadow is shortest. Because the sun is highest in the sky at midday, so light comes from almost straight above and makes a short shadow.
2. Yes, the shadow moved. The shadow moves because Earth is rotating (spinning), which makes the sun appear to be in a different position in the sky.
3. **False.** The sun does not actually move — it only appears to move.
4. *Earth* is actually moving (rotating/spinning).

⭐ Think Like a Scientist — Sample Answer

With a flashlight at night, you could shine it from one side (morning), above (midday), and the other side (afternoon) and get similar shadow patterns. The flashlight represents the sun. The pattern would be the same because the light source position determines shadow direction.

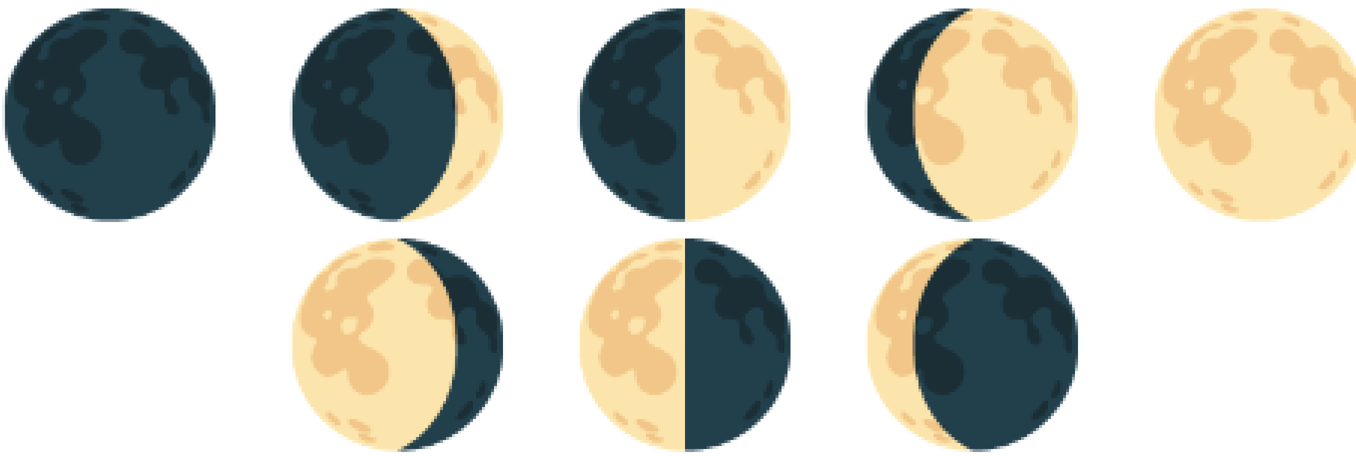
✅ What Success Looks Like

- Child can describe that shadows changed direction and length throughout the day
- Child understands that the shortest shadow occurred at midday (when the sun was highest)
- Child can state that *Earth* rotates, making the sun appear to move
- Child followed the sun safety rule throughout the activity

🌙 The Moon — Earth's Natural Satellite

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 2 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes + ongoing nightly observations 🔬 **Subject:** Earth Science — Sun, Moon & Stars 🏠 **Materials:** Household only



The moon moves through a full cycle of phases every 29.5 days

📋 **NGSS Standard: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Observe* and *record* the moon's apparent shape over several days
- *Describe* the pattern of moon phases and predict what comes next
- *Explain* why the moon appears to change shape (we see different portions of the lit side)

🔬 The Big Science Idea

The moon seems to change shape every night — but it doesn't actually change! The moon is always a sphere (a ball shape). What changes is *how much of the lit side we can see* from Earth.

The moon orbits (travels around) Earth once every 29.5 days. The sun always lights up one half of the moon — but as the moon moves around Earth, our viewing angle changes. Sometimes we see most of the lit half (full moon), sometimes we see just a sliver (crescent), and sometimes the lit side faces entirely away from us (new moon). All the shapes in between are the different phases.

This creates a predictable, repeating pattern — one of the most reliable patterns in all of nature!

> 💡 **Key vocabulary:** moon, phase, orbit, satellite, reflect, crescent, quarter, gibbous, full moon, new moon, cycle

🌑
New Moon
Not visible — lit side faces away from Earth

🌒
Crescent
A thin curved sliver of light

🌓
First Quarter
Half of the moon lit up

🌔
Gibbous
More than half lit up

🌕
Full Moon
Whole lit side faces Earth — brightest!

🌖
Gibbous
Still more than half lit, fading

🌗
Last Quarter
Half lit again — other half this time

🌘
Crescent
Thin sliver before new moon returns

📦 Materials

🏠 What You'll Need

- This Moon Journal (printed — Student Worksheet page)
- Crayons or pencil for drawing moon shapes
- A clear night sky — or a moon phase calendar/chart (if sky is cloudy)

Timing tip: Starting this journal on or just after a new moon gives students the most dramatic arc — they'll watch the moon grow from invisible to full over two weeks. Check a moon phase calendar at ***timeanddate.com*** *to plan your start date.*

🔬 The Moon Journal Activity

Step 1 — Introduce the Moon (Day 1, in class)
Look at the moon phase diagram together. Ask: *"Has anyone ever noticed the moon looks different on different nights?"* Show the eight phases and introduce their names. Explain: the moon isn't actually changing shape — we're just seeing different amounts of the lit half.

Step 2 — Moon Model (Day 1, in class)
Take a flashlight and a tennis ball into a dark room. Shine the flashlight on the ball and slowly walk in a circle around it. Watch how different amounts of the lit side face you at different points. Explain: *"This is how moon phases work. The moon is always lit by the sun, but from Earth we see different amounts of the lit half."*

Step 3 — Start the Journal (Nights 1–14)
Each night before bedtime, go outside (or look out a window) and find the moon. Draw what you see in that night's box on the Moon Journal. If the moon isn't visible yet, write "not up yet" or "cloudy." If there's a new moon, draw a blank circle and write "new moon."

Step 4 — Name the Phase
After drawing, look at the moon phase chart and write the name of the phase next to your drawing. Does it match what you drew?

Step 5 — Mid-Point Check-In (Day 7)
After a week of observations, look at your journal so far. Ask: *"Is the moon getting bigger (waxing) or getting smaller (waning)?"* Make a prediction: what will it look like in 7 more days?

Step 6 — Final Reflection (Day 14+)
Review the complete journal. Fill in the reflection questions on the worksheet. Can you now predict what phase comes next?

💡 **Can't see the moon?** Cloudy nights happen! Keep a note ("cloudy" or "foggy") rather than skipping the entry. Comparing what you expected vs. what you actually saw is part of real scientific observation.

💬 Discussion Questions

🗣️ **Talk about these together:**

- What do you notice about the pattern? Does the moon follow a predictable order?
- If the moon is a crescent tonight, what will it look like in about two weeks?
- Why can we sometimes see the moon during the day?
- Has anyone ever noticed the moon looks bigger near the horizon? (This is the "moon illusion" — a trick our brains play!)

🌟 **Big Question: "Does the moon make its own light?"**

No — the moon has no light of its own. It reflects sunlight, like a mirror. That's why it's only visible when sunlight hits its surface and bounces toward Earth. The "dark side of the moon" isn't actually dark — it receives just as much sunlight as the near side. It's just always facing away from Earth. When we say "dark side" we mean the side we never see, not a permanently dark side.

🚀 Extensions

- 📅 **Full Cycle:** If you started with a new moon, continue the journal for a full 29-day cycle and see the moon return to new.
- 📚 **Read Aloud:** *"The Moon Book"* by Gail Gibbons or *"Papa, Please Get the Moon for Me"* by Eric Carle
- 🌍 **Fun Fact:** The moon is slowly moving away from Earth at about 3.8 centimeters per year — about the rate your fingernails grow!

👩‍🏫 Parent/Teacher Notes

- **This is a long-form observation activity:** The Moon Journal works best over 7–14 days. Build it into the bedtime routine — kids love having a reason to look at the sky each night.
- **Moon visibility:** The moon isn't always visible at bedtime. Around new moon, it sets soon after the sun and won't be visible in the evening sky. Check a moon phase app so you know what to expect.
- **Common misconception:** Many children think a lunar eclipse causes the phases. Clarify: phases happen every month because of the moon's orbit; eclipses are rare events when Earth's shadow falls on the moon.
- **The "waxing/waning" vocabulary:** Waxing = growing (getting more lit up). Waning = shrinking (getting less lit up). These are real astronomy terms and first-graders can absolutely learn them.
- **If cloudy for most of the 2 weeks:** Use a monthly moon phase chart or an app like "Moon" to show what the moon looks like — then have the child draw from the chart instead of direct observation.

🌙 Student Worksheet — Moon Journal

Unit 1, Lesson 2 — The Moon

Name: ________________________________ Start Date: ________________

Each night, go look at the moon and draw what you see in that day's box. Write the phase name below your drawing!

Night 1 Date: ______ Phase: _____	**Night 2** Date: ______ Phase: _____	**Night 3** Date: ______ Phase: _____	**Night 4** Date: ______ Phase: _____	**Night 5** Date: ______ Phase: _____	**Night 6** Date: ______ Phase: _____	**Night 7** Date: ______ Phase: _____
Night 8 Date: ______ Phase: _____	**Night 9** Date: ______ Phase: _____	**Night 10** Date: ______ Phase: _____	**Night 11** Date: ______ Phase: _____	**Night 12** Date: ______ Phase: _____	**Night 13** Date: ______ Phase: _____	**Night 14** Date: ______ Phase: _____

✏️ Reflection Questions

1. Did the moon follow a pattern? (circle) **YES** / **NO** — Describe what you noticed: ______________________________

2. The moon got (circle one): **bigger then smaller** / **smaller then bigger** / **stayed the same**

3. Does the moon make its own light? (circle) **YES** / **NO** — Where does moonlight come from? _________________

4. Draw what you predict the moon will look like in 2 weeks:

✂️ PARENT ANSWER KEY — Detach before giving worksheet to students

🌙 Moon Journal — What to Expect

The specific phases your child observes will depend on when they start. The general cycle: new moon (not visible) → waxing crescent → first quarter (half lit) → waxing gibbous → full moon → waning gibbous → last quarter → waning crescent → new moon again. The full cycle takes 29.5 days.

✏️ Reflection Answers

1. Yes, the moon does follow a pattern. It follows a predictable cycle of phases that repeats every 29.5 days.
2. The moon gets *bigger* (waxing — more lit each night) then *smaller* (waning — less lit each night). Then the cycle repeats.
3. No, the moon does not make its own light. Moonlight is sunlight reflecting off the moon's surface.
4. In 2 weeks: if starting from new moon → should be near full moon. If starting from full moon → should be near new moon (very thin or invisible).

💡 Moon Phase Quick Reference

- 🌑 **New Moon** — Moon is between Earth and Sun; lit side faces away. Not visible at night.
- 🌒🌘 **Crescent** — Less than half lit. Waxing crescent (right side lit) → growing. Waning crescent (left side lit) → shrinking.
- 🌓🌗 **Quarter** — Exactly half lit. First quarter is waxing; last quarter is waning.
- 🌔🌖 **Gibbous** — More than half lit. Waxing gibbous → growing toward full. Waning gibbous → shrinking from full.
- 🌕 **Full Moon** — Earth is between Sun and Moon; we see the fully lit side.

✅ What Success Looks Like

- Child can name at least 4 moon phases and put them in order
- Child understands the moon follows a repeating pattern (~29.5 days)
- Child can explain that the moon reflects sunlight rather than making its own light
- Child can predict what phase will come next given a current phase

Little Lab Coats

⭐ Stars — Suns Very Far Away

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 3 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Earth Science — Sun, Moon & Stars 🏠 **Materials:** Household only

🌟✨⭐🌟✨

Every star you see at night is a sun — just unimaginably far away

📋 **NGSS Standard: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Explain* why stars are only visible at night (and why the sun makes them invisible during the day)
- *Describe* observable patterns of stars in the sky, including recognizable groupings
- *Identify* the Big Dipper and three other common star patterns using picture cards and a simple star chart

🔬 The Big Science Idea

Here's a surprising fact: stars are shining in the sky *right now* — even during the day! We just can't see them because our sun is so bright it outshines everything else. When the sun sets and the sky darkens, the stars become visible again — it's like turning off a bright light in a room and suddenly being able to see the stars painted on the ceiling.

Every single star you see is a sun — some smaller and dimmer than ours, many far larger and brighter. They only look tiny and faint because they are *unimaginably far away*. The closest star besides our sun (Proxima Centauri) is so far that traveling to it at the speed of an airplane would take about 4 million years.

Even though stars seem scattered randomly, some form recognizable patterns called *constellations* and *asterisms*. The Big Dipper is one of the most reliable and easiest to find — like a ladle or a dipper scoop shape in the northern sky.

💡 **Key vocabulary:** star, constellation, asterism, Big Dipper, light-year, visible, atmosphere, night sky

📦 Materials

🏠 What You'll Need

- A flashlight
- A bright room (normal room lighting)
- A dark room or closet
- This lesson's constellation picture cards and tracing worksheet
- Crayons or pencil

For bonus nighttime activity: a clear night, your yard or a window, and the Big Dipper star chart to find the real thing!

🔬 The Flashlight Experiment — Why Can't We See Stars During the Day?

🔦 What You Need

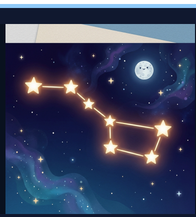
A flashlight and two different rooms: one bright, one dark.

Step 1 — The Bright Room Test
Turn on the flashlight in a normally lit room (lights on, sunshine through windows). Point it at the wall. Can you see its beam? Can you clearly see its light on the wall? Ask: *"Is the flashlight on?"* Yes, but it's hard to notice because the room is already bright.

Step 2 — The Dark Room Test
Move to a dark room — close the curtains, turn off the lights. (A closet works great!) Turn on the same flashlight and point it at the wall. Wow — now it's very easy to see! Ask: *"Did the flashlight get brighter?"* No! The flashlight is the same. The difference is the darkness around it.

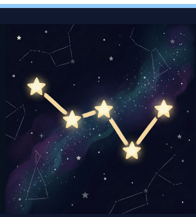
Step 3 — Connect It to Stars
Explain: Stars are like that flashlight. They're always shining. But during the day, our sun is so incredibly bright that it lights up the whole sky — just like the bright room made the flashlight hard to see. At night, the sun is on the other side of Earth, so the sky gets dark — just like the dark room — and suddenly all those flashlights (stars) become visible!

Step 4 — Spot Four Constellations (Nighttime Activity)
On a clear night, take the constellation cards outside and try to spot all four patterns over time: the **Big Dipper**, **Orion**, **Cassiopeia**, and **Scorpius**. Start by matching what you see in the sky to the card pictures. Some constellations are easier in different seasons, so treat this as an ongoing spotting challenge rather than a one-night test.

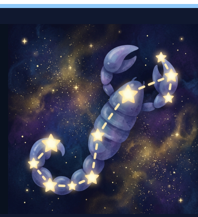


Big Dipper

Orion



Cassiopeia



Scorpius

💡 **Constellation spotting tip:** The Big Dipper and Cassiopeia are strong northern-sky patterns. Orion is a great winter constellation because of its clear three-star belt. Scorpius is a great summer pattern with a long curved body. You may not see all four in one season, so keep the cards and return to them across the year.

💬 Discussion Questions

🗣️ **Talk about these together:**

- Why couldn't you see the flashlight as well in the bright room?
- If you were on the moon, would the sky be black even during the "day"? (Yes! The moon has no atmosphere to scatter sunlight, so the sky is black. Stars are there to see — though astronauts' eyes were adapted to the bright surface, making faint stars hard to notice in person.)
- Why do stars twinkle? (Hint: Earth's atmosphere makes starlight shimmer — like looking at a penny through a stream of water.)
- Can you think of any other reason why stars might be harder to see? (Light pollution from cities!)

🌟 **Big Question: "Stars look tiny from Earth but the sun looks huge. Is the sun special?"**

The sun is actually a pretty average star — not especially big or bright by stellar standards. It looks huge and bright only because it's incomparably closer: about 93 million miles away. The next nearest star (Proxima Centauri) is about 4.2 light-years away — so far that its light takes 4.2 years to reach us. If the sun were the size of a basketball, the next nearest star would be about 5,000 miles away.

🚀 Extensions

- 🌃 **Dark Sky Adventure:** Drive away from city lights to see many more stars. Areas far from cities can show thousands of stars — and even the Milky Way!
- 📱 **Star App:** Free apps like "Star Walk Kids" or "SkyView Lite" let you point your phone at the sky and identify stars and constellations.
- 🌐 **Print a Star Chart:** Visit *stellarium.org* and print a free star map for your location and current date.
- 📚 **Read Aloud:** *"Zoo in the Sky"* by Jacqueline Mitton, or *"Astrophysics for Young People in a Hurry"* by Neil deGrasse Tyson (simplified for adults to read aloud)
- 🎯 **Constellation Stories:** Every culture has stories for star patterns. Look up the Greek myth for Orion or the Native American stories for the Big Dipper.

👩‍🏫 Parent/Teacher Notes

- **The flashlight demo is the heart of this lesson:** The "bright room vs. dark room" observation is memorable and directly maps to the concept. Let kids do it themselves — don't just demonstrate it for them.
- **Stars are always there:** Emphasize this gently — it's a profound idea. The stars you see at night are the *same* stars that exist during the day. We just can't see them. This can spark genuine wonder.
- **Big Dipper vs. constellation:** Technically the Big Dipper is an *asterism* (a recognizable pattern of stars), not a true constellation. It's part of the constellation Ursa Major. For Grade 1, "star pattern" or "star shape" is perfectly fine.
- **The North Star:** If kids find the Big Dipper, show them how the two stars at the outer edge of the cup point to Polaris (the North Star). This is a real navigation trick that sailors used for centuries.
- **Light pollution discussion:** Kids in cities may have never seen a truly dark sky. This is a meaningful discussion — artificial light affects both humans and wildlife.

⭐ Student Worksheet — Stars & the Big Dipper

Unit 1, Lesson 3 — Stars

Name: ________________________________ Date: ________________

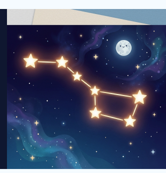
🔦 The Flashlight Experiment

Fill in your observations from the flashlight test!

Test	Could you see the flashlight well?	Why?
Bright Room		
Dark Room		

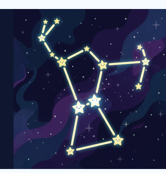
⭐ Constellation Cards and Tracing

Study these four common constellations. Trace the star shapes with your finger first. Then decide if the pattern looks anything like its name!



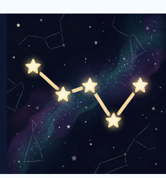
Big Dipper

Trace the shape and say what it looks like.



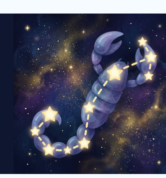
Orion

Trace the shape and circle Orion's belt.



Cassiopeia

Trace the W shape.



Scorpius

Trace the curved tail shape.

✏️ Answer These Questions

1. Are stars visible during the day? (circle) **YES** / **NO** — Why? ______________________

2. Why did the flashlight look brighter in the dark room? ______________________

3. The flashlight stands for the ______________ in our experiment. The bright room stands for ______________. The dark room stands for ______________.

4. True or False: The sun is a very special, one-of-a-kind kind of star. (circle) **TRUE** / **FALSE**

🌃 Nighttime Challenge

Go outside on clear nights and try to spot all four constellation patterns over time. Check the boxes as you find them:

☐ I went outside and looked at the night sky

☐ I spotted the Big Dipper

☐ I spotted Orion

☐ I spotted Cassiopeia

☐ I spotted Scorpius

Draw what the night sky looked like:

✂️ PARENT ANSWER KEY — Detach before giving worksheet to students

🔦 Flashlight Experiment Answers

- **Bright Room:** Could *not* see the flashlight well. Because the room was already bright, so the flashlight light blended in with the background light.
- **Dark Room:** *Could* see the flashlight very well. Because there was no competing light, so the flashlight's light stood out clearly.

✏️ Worksheet Answers

1. **No**, stars are not visible during the day. The sun's light is so bright that it overwhelms the light from distant stars. The stars are still there — we just can't see them.
2. The flashlight looked brighter in the dark room because there was no other light competing with it. The flashlight itself didn't change — only the environment around it changed.
3. The flashlight stands for a *star*. The bright room stands for *daytime (with the sun)*. The dark room stands for *nighttime*.
4. **False.** The sun is actually a fairly average star. It only seems special because it's much, much closer to us than any other star.

🌠 Constellation Notes

The Big Dipper, Orion, Cassiopeia, and Scorpius are four memorable constellation patterns for children to study over time. The Big Dipper and Cassiopeia are strong northern-sky patterns. Orion is especially easy because of its clear three-star belt. Scorpius has a long curved body that makes it stand out in summer skies.

✅ What Success Looks Like

- Child can explain why stars aren't visible during the day (the sun's light is too bright, not because the stars turn off)
- Child understands that stars are always present — just hidden by sunlight
- Child can identify or describe four common constellation patterns using the cards and tracing page
- Child understands that the sun is just a star that looks bigger because it's much closer

Little Lab Coats

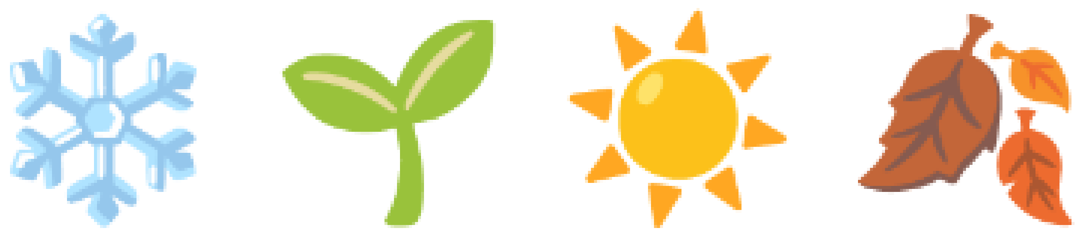
📅 How Much Daylight? Seasons and the Sun

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 4 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Earth Science — Sun, Moon & Stars 🏠 **Materials:** Household only

⚠️ Sun Safety Reminder

This lesson involves observing daylight patterns — not looking at the sun. As always, **never look directly at the sun.** Daylight observations mean noticing when it's light or dark outside — keep eyes away from the sun itself.



Winter, Spring, Summer, Fall — daylight hours change dramatically across the seasons

📋 **NGSS Standard: 1-ESS1-2** — Make observations at different times of year to relate the amount of daylight to the time of year.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Compare* the hours of daylight across the four seasons
- *Explain* the pattern: more daylight in summer, less in winter
- *Identify* Earth's tilt (not distance from the sun) as the cause of seasons
- *Create* a bar graph showing seasonal daylight changes

🔬 The Big Science Idea

Have you ever noticed that summer days last longer and winter days seem short? This is real! The number of daylight hours changes throughout the year in a predictable pattern.

In summer, the sun rises early, climbs high in the sky, and sets late — giving us more hours of light. In winter, the sun rises later, stays lower in the sky, and sets early — giving us fewer hours of light.

Here's the surprising part: this has *nothing* to do with how close Earth is to the sun! Earth is actually slightly *closer* to the sun in January (winter in the Northern Hemisphere) than in July (summer). The cause of seasons is Earth's *tilt*.

Earth spins at a tilt of about 23.5 degrees. In summer, our hemisphere is tilted *toward* the sun — days are longer and sunlight hits us more directly (more energy, more warmth). In winter, our hemisphere is tilted *away* — days are shorter and sunlight hits us at a glancing angle (less energy, less warmth).

💡 **Key vocabulary:** season, daylight, sunrise, sunset, tilt, hemisphere, pattern, summer solstice, winter solstice, equinox

☀️ **Sample Daylight Hours (Denver, CO — approximate):**
Winter solstice (Dec 21): ~9 hours 21 minutes of daylight
Spring equinox (Mar 20): ~12 hours 10 minutes of daylight
Summer solstice (Jun 21): ~14 hours 58 minutes of daylight
Fall equinox (Sep 22): ~12 hours 10 minutes of daylight
*Look up your location's actual times at **timeanddate.com/sun***

📦 Materials

🏠 What You'll Need

- Paper and crayons
- Ruler (to draw the bar graph)
- A flashlight or lamp
- A ball, orange, or small globe to act as Earth
- 4 strips of paper (or sentence strips) for comparing daylight lengths
- Sunrise/sunset data — parent looks up for your location at *timeanddate.com/sun* or a weather app
- This worksheet (printed)

🔬 The Daylight Data Activity

Step 1 — Act Out the Seasons
Start with a quick movement model. Put a lamp or flashlight in the middle of the room as the sun. Have your child hold a ball, orange, or small globe as Earth and keep it tilted the same way the whole time. Slowly walk around the light and stop at "summer" and "winter." Ask: *"Is our side getting more light or less light now?"* Keep this very simple. The goal is just to notice that the tilt changes how much light our side gets.

Step 2 — Try a Flashlight Demo
Shine the flashlight at the tilted ball. Show how the light hits more directly in summer and at more of a slant in winter. Let your child turn the ball and point to the part getting the most light. Then connect it back to daylight: when our side is tilted toward the sun, we get longer days.

Step 3 — Gather the Data
Parent: Before or during the lesson, look up the sunrise and sunset times for your location for four dates:

- Around **December 21** (winter solstice — shortest day)
- Around **March 20** (spring equinox — equal day/night)
- Around **June 21** (summer solstice — longest day)
- Around **September 22** (fall equinox — equal day/night)

Write the sunrise and sunset times in the Data Table on the worksheet.

Step 4 — Build Daylight Strips and Graph the Pattern
First, make four paper strips for winter, spring, summer, and fall. Cut or compare the strips so winter is shortest, summer is longest, and spring and fall are in between. Line them up from shortest to longest. Then use the same information to color in the bar graph on the worksheet. Use different colors for each season.

Step 5 — Read the Pattern
Look at the completed strips and graph together. Ask: *"Which season has the most daylight? Which has the least? What pattern do you see as the year goes from winter to summer and back again?"*

Step 6 — Check Real Life Outside
If possible, step outside and notice where shadows are or talk about a time earlier in the day when shadows looked different. Connect the idea back to the lesson: the sun's path changes across the year, and that helps create different amounts of daylight in different seasons.

💬 Discussion Questions

🗣️ **Talk about these together:**

- Which season did you enjoy having more daylight in? Why?
- How would your day be different if the sun set at 4pm vs. 9pm?
- People often say "in summer we're closer to the sun." Do you think that's true? (Actually, we're closer to the sun in winter!)
- Can you think of animals that might behave differently based on how much daylight there is? (Bears hibernating, birds migrating!)

🌟 **Big Question: "In Alaska in summer, the sun doesn't set for weeks. How can people sleep?"**

This is called the Midnight Sun and it happens above the Arctic Circle. The tilt is so extreme that the sun stays above the horizon 24 hours a day for weeks. People use blackout curtains, eye masks, and adjust their sleep schedules. In winter, the opposite happens — the sun doesn't rise for weeks (Polar Night). Residents report it affects mood and energy levels significantly, a condition called Seasonal Affective Disorder (SAD).

🚀 Extensions

- 🌍 **Flip the Seasons:** When it's summer in the Northern Hemisphere, it's winter in Australia (Southern Hemisphere). Use the flashlight and ball again, but point to the opposite side of Earth to show why.
- 📏 **Daylight Strip Match:** Make new paper strips for different months and have your child put them in order from shortest daylight to longest daylight.
- 👣 **Shadow Walk:** Go outside in the morning and later in the afternoon and compare shadow length and direction. Talk about how the sun's position changes through the day.
- 🌱 **Plant and Animal Connection:** Talk about animals, flowers, or bedtime routines that change when days are longer or shorter.

👩‍🏫 Parent/Teacher Notes

- **Seasons ≠ distance from sun:** This is one of the most common misconceptions in science. Earth is slightly closer to the sun in January. The seasons are caused by tilt, not distance. This is worth spending extra time on.
- **The "solstice" words:** "Solstice" means "sun stands still" in Latin — at the solstices, the sun's position at noon stops getting higher or lower for a few days. "Equinox" means "equal night" — day and night are roughly equal in length.
- **Hemisphere specifics:** This lesson describes the Northern Hemisphere (where most US families live). If you're in the Southern Hemisphere, the seasons are reversed — December is summer, June is winter.
- **Grade 1 depth:** For this age, the key takeaways are: (1) daylight hours change through the year, (2) summer has more, winter has less, (3) this creates a predictable repeating pattern. Keep the tilt explanation very concrete with the flashlight and ball model.
- **Make it hands-on:** The strongest version of this lesson is movement first, graph second. Let the child hold the "Earth," walk around the "sun," and build paper daylight strips before doing the worksheet.
- **The bar graph:** Grade 1 students may need help drawing the bars accurately. It's fine for the parent to draw the axes and label them while the child colors in the bars and observes the pattern.

📅 Student Worksheet — Daylight Bar Graph

Unit 1, Lesson 4 — Daylight and Seasons

Name: ________________________________ Date: ________________

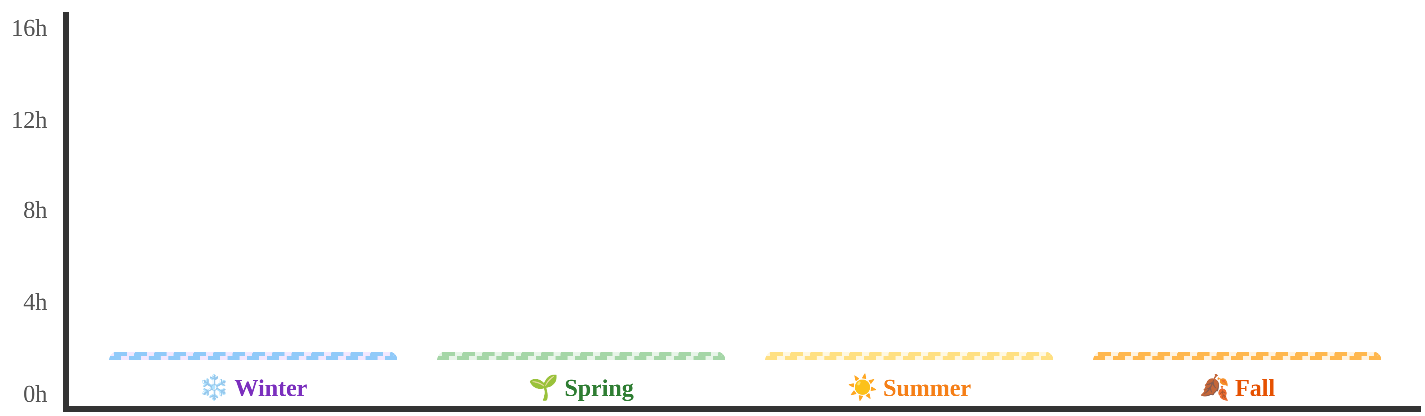
📊 Daylight Data Table

Fill in the sunrise, sunset, and total daylight hours for each season. Your parent will help you find the data!

Season	Date Used	Sunrise Time	Sunset Time	Total Hours of Daylight
❄️ **Winter**	Dec 21			
🌱 **Spring**	Mar 20			
☀️ **Summer**	Jun 21			
🍂 **Fall**	Sep 22			

📊 Color the Bar Graph

Color each bar up to the number of daylight hours for that season. Use a different color for each season! The Y-axis goes from 0 to 16 hours.



Color each bar from the bottom up to show the hours of daylight for that season!

✏️ Answer These Questions

1. Which season has the *most* hours of daylight? ______________________
2. Which season has the *fewest* hours of daylight? ______________________
3. Do the hours of daylight follow a pattern? (circle) **YES** / **NO**

4. What causes the seasons? (circle the correct answer):

- ☐ Earth gets closer to the sun in summer
- ☐ Earth tilts toward or away from the sun
- ☐ The sun gets bigger in summer

✂️ PARENT ANSWER KEY — Detach before giving worksheet to students

📊 Sample Daylight Data (Denver, CO)

- Winter (Dec 21): Sunrise ~7:17am, Sunset ~4:38pm → **~9 hours 21 min**
- Spring (Mar 20): Sunrise ~6:21am, Sunset ~6:31pm → **~12 hours 10 min**
- Summer (Jun 21): Sunrise ~5:31am, Sunset ~8:29pm → **~14 hours 58 min**
- Fall (Sep 22): Sunrise ~6:38am, Sunset ~6:48pm → **~12 hours 10 min**

Look up your specific location at timeanddate.com/sun for accurate local times.

✏️ Worksheet Answers

1. **Summer** has the most hours of daylight (~15 hours at the summer solstice in Denver).
2. **Winter** has the fewest hours of daylight (~9 hours at the winter solstice in Denver).
3. **Yes** — the hours follow a predictable pattern that repeats every year.
4. The correct answer is: *Earth tilts toward or away from the sun.* Earth's tilt causes seasons — not its distance from the sun.

📈 What the Graph Should Show

The bar graph should show: Winter (shortest bar) → Spring (medium-tall) → Summer (tallest bar) → Fall (medium-tall, similar to Spring). The pattern looks like a hill: it rises from winter to summer, then falls again toward winter.

✅ What Success Looks Like

- Child can read the bar graph and identify which season has more or less daylight
- Child can describe the seasonal pattern (more daylight in summer, less in winter)
- Child understands this pattern repeats every year (predictable)
- Bonus: Child can explain that Earth's tilt causes seasons, not closeness to the sun

Little Lab Coats

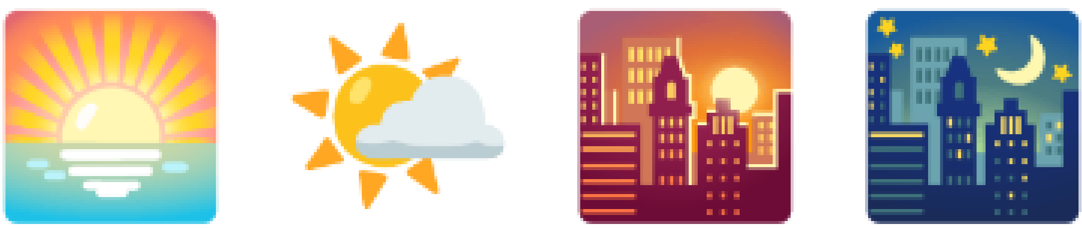
🌅 The Sky Changes — Morning, Day & Night

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 5 of 6

📚 **Grade:** 1st Grade
⏱️ **Duration:** 45 minutes + brief observations across one day
🔬 **Subject:** Earth Science — Sun, Moon & Stars
🏠 **Materials:** Household only

⚠️ Sun Safety Reminder

When observing the sky at sunrise or during the day, **always look at the sky around the sun — not directly at the sun itself.** You can observe sky colors, clouds, and position of light without ever looking at the sun directly. Keep this rule all day!



Sunrise, midday, sunset, night — the sky transforms completely across a single day

📋 **NGSS Standard: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Observe* and *record* how the sky's appearance changes from morning to night
- *Identify* patterns in sky color, light level, and object visibility at different times
- *Explain* why the sky is blue during the day and red/orange at sunrise and sunset

🔬 The Big Science Idea

The sky is never the same twice — it changes constantly through the day. But these changes follow a *predictable pattern* that repeats every single day!

🌅 **Sunrise:** The sky glows orange, pink, and red near the horizon. The sun rises in the east. Stars fade away as the sky brightens.
🌤️ **Midday:** The sky is bright blue. The sun is highest. Shadows are shortest. No stars visible.
🌇 **Sunset:** The sky turns orange, red, and purple again near the horizon. The sun sets in the west.
🌃 **Night:** The sky is dark. Stars appear as Earth rotates away from the sun. The moon may be visible.

Why does the sky change color? It's all about how sunlight travels through Earth's atmosphere. Sunlight contains *all the colors of the rainbow*. When sunlight enters the atmosphere, gas molecules scatter the shorter (blue) wavelengths in all directions — this scattered blue light is what makes the sky look blue during the day.

At sunrise and sunset, sunlight has to travel through much more atmosphere to reach your eyes (because the sun is near the horizon). By the time it arrives, most of the blue light has already scattered away. What's left are the longer red and orange wavelengths — giving us those gorgeous pink and orange skies!

💡 **Key vocabulary:** atmosphere, scatter, wavelength, Rayleigh scattering, sunrise, sunset, observation, pattern

🌈 Why the Sky Changes Color — A Visual

Sunrise	Midday	Sunset	Night
🌅	🌤️	🌇	🌃

At sunrise and sunset, light travels through more atmosphere → blue scatters away → orange/red remains. At midday, light travels through less atmosphere → blue dominates.

📦 Materials

🏠 What You'll Need

- This Sky Journal worksheet (printed)
- Crayons (especially: blue, orange, yellow, pink, red, black, white)
- A pencil for notes
- A window with a view of the sky — or ability to briefly go outside at 4 different times

Best timing: Plan for a day when you can look at the sky right at sunrise/early morning, around midmorning (9–10am), afternoon (2–3pm), and after dark. Even just looking out a window works fine!

🔬 The Sky Journal Activity

Step 1 — Sunrise / Early Morning Observation
Go to a window or outside just before or just after sunrise. *Look at the sky — not the sun!* Notice: What color is the sky? What colors are near the horizon? Are there any clouds? Can you still see stars fading? Is the moon visible? Draw and record what you see.

Step 2 — Midmorning Observation (9–10am)
Look at the sky again around mid-morning. How has it changed from sunrise? The sky should be bright blue by now. Where is the sun positioned in the sky (low, medium, high)? Record your observations.

Step 3 — Afternoon Observation (2–3pm)
Look at the sky in the early afternoon. Is the sky still blue? Are there more or fewer clouds than before? Is the sun starting to lower in the west? Record your observations.

Step 4 — After Dark Observation
Once it's fully dark, go outside or look out a window. What do you see? Stars? The moon? How many stars can you count? Which direction is the moon? Record and draw what you see.

Step 5 — Compare and Find Patterns
Look at all four drawings together. What changed? What stayed the same? What pattern did you notice? Fill in the reflection questions on the worksheet.

💬 Discussion Questions

🗣️ Talk about these together:

- Which time of day was your favorite sky? Why?
- What was the biggest difference between the sunrise sky and the midday sky?
- Can you predict what the sky will look like tomorrow at the same times? (Yes — because it follows a pattern!)
- Why do you think sunsets look different depending on whether there are clouds or not?

🌟 Big Question: "Why is the sky blue during the day but orange and red at sunrise and sunset?"

Sunlight contains all colors. When light travels through the atmosphere, gas molecules scatter shorter (blue) wavelengths in all directions — that scattered blue light is what we see as the blue sky. At sunrise and sunset, sunlight travels through much more atmosphere to reach your eyes. By then, most of the blue light has scattered away, leaving the longer red and orange wavelengths. So the same sunlight looks different depending on how much atmosphere it travels through.

🚀 Extensions

- 🌈 **Cloud Types:** While observing the sky, look up what different cloud types look like (cumulus, stratus, cirrus). Can you spot them during your observations?
- 📸 **Photo Journal:** Take a photo of the sky at each of the four times instead of (or in addition to) drawing. Compare the photos side by side!
- 🌊 **Rayleigh Scattering Demo:** Fill a clear glass with water and add a tiny drop of milk. Shine a flashlight through the side — the water looks bluish from the side (scattered blue light) but the transmitted light looks orange-red (like a sunset). This is the same effect!
- 🌍 **Other Planets:** On Mars, the sky is pinkish-tan because the atmosphere is mostly carbon dioxide with red dust particles. The sky color is determined by the atmosphere's composition!

👩‍🏫 Parent/Teacher Notes

- **Safety at sunrise/sunset:** When looking at orange and pink sky colors near the horizon, the sun may be very low and close to the field of view. Always remind the child to keep their eyes away from the sun itself — look at the sky color *around* it, not at the sun.
- **Rayleigh scattering:** This is the scientific term for why the sky is blue. Don't worry about having children memorize it — but if they're curious, the concept is "smaller molecules scatter shorter (blue) light, longer (red/orange) light passes through more." The milk-in-water demonstration is very effective at making this concrete.
- **Night sky timing:** How long after sunset before it's truly dark varies by season and latitude. In summer, it may be 9pm or later before stars are visible. Check dusk/dark times online if needed.
- **What if the day is cloudy?** Cloudy days work too! Have the child observe: What color is a cloudy sky vs. a clear sky? Why does overcast sky look gray? (Clouds scatter all wavelengths equally — so the light appears white/gray.)
- **The science depth:** For Grade 1, the key observations are: sky changes color throughout the day, there's a predictable pattern, and we can describe what those changes are. The Rayleigh scattering explanation is for curious families who want to go deeper.

🌅 Student Worksheet — Sky Journal

Unit 1, Lesson 5 — The Sky Changes

Name: ______________________________ Date: _______________

Look at the sky 4 times today! Draw what you see and fill in the notes for each observation.

🌅 Sunrise / Early Morning

Sky color to try: orange, pink, yellow, purple

Draw the sky here

Sky color: ________________

Clouds? ________________

Stars still visible? ________________

🌤️ Midmorning (9–10am)

Sky color to try: blue, white for clouds, yellow sun

Draw the sky here

Sky color: ________________

Sun position (low/middle/high): ________

Clouds? ________________

🌇 Afternoon / Sunset

Sky color to try: orange, red, deep blue, gold

Draw the sky here

Sky color: ________________

Sun position (low/middle/high): ________

Different from morning? ________________

🌃 After Dark (night)

Sky color to try: dark blue, black, yellow stars, white moon

Draw the sky here

Moon visible? ________________

Stars visible? How many? ________

Sky color: ________________

✏️ Reflection Questions

1. What was the biggest change you noticed between morning and night? ____________________
2. When was the sky *blue*? When was it *orange or red*? ____________________

3. Do you think tomorrow's sky will follow the same pattern? (circle) **YES** / **NO** — Why? ______________________

4. In your own words, why is the daytime sky blue? ______________________

✂️ PARENT ANSWER KEY — Detach before giving worksheet to students

🌅 Sky Journal — What to Expect

- **Sunrise/Early Morning:** Sky near the horizon will be orange/pink/red (or gray if cloudy). Stars may still be fading. Moon may be visible if it was a full or partial phase the night before.
- **Midmorning:** Sky is bright blue (clear day) or gray-white (overcast). Sun is medium height in the eastern to southern part of the sky. Stars not visible.
- **Afternoon/Sunset:** Sky begins to shift orange/gold near the horizon as the sun lowers. Blue sky still overhead. Stars appear as the sky darkens after sunset.
- **Night:** Dark sky with stars visible (if clear). Moon may be visible depending on phase. No blue sky visible.

✏️ Reflection Answers

1. Biggest change: color of the sky (blue during the day, dark at night with stars visible). Also: the sun moves across the sky and disappears, then stars appear.
2. The sky was blue *during the day* (midmorning through afternoon). It was orange or red *at sunrise and sunset*.
3. **Yes** — the sky follows the same daily pattern predictably. Every day follows the same sequence (sunrise colors → blue day → sunset colors → dark night).
4. The daytime sky is blue because sunlight has all colors, and the atmosphere scatters blue light in all directions, making the whole sky appear blue. (Rayleigh scattering — great if they know this term!)

🌈 Rayleigh Scattering — Parent Background

Lord Rayleigh (John William Strutt) described in 1871 why the sky is blue. Gas molecules in the atmosphere are much smaller than the wavelength of visible light. These tiny molecules interact more strongly with shorter wavelengths (blue/violet). This is Rayleigh scattering. Our eyes are slightly less sensitive to violet, and there's more blue in sunlight than violet, so we perceive the scattered light as blue.

✅ What Success Looks Like

- Child can describe at least 3 distinct sky appearances (morning, midday, night)
- Child notices the pattern repeats daily and is therefore predictable
- Child can explain that sky color changes because of how sunlight travels through the atmosphere
- Child can describe when stars become visible and why (sun on other side of Earth → sky darkens)

Little Lab Coats

🔭 Patterns in the Sky

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 6 of 6 ⭐ Final Lesson

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Earth Science — Sun, Moon & Stars 🏠 **Materials:** Household only

The sun, moon, and stars — three cosmic objects with patterns scientists can predict for centuries

📋 **NGSS Standards: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted. | **1-ESS1-2** — Make observations at different times of year to relate the amount of daylight to the time of year.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Summarize* the observable patterns of the sun, moon, and stars across different time periods
- *Predict* what the sky will look like at a given time based on known patterns
- *Connect* all unit concepts — daily sun motion, moon phases, star visibility, daylight seasons, sky color changes

🔬 What We've Learned: Sky Patterns Summary

Object	Pattern Over a Day	Pattern Over a Month	Pattern Over a Year
☀️ **Sun**	Rises east, arcs across sky, sets west. Shadow moves and changes length.	Same daily pattern repeats	Higher in summer (more daylight), lower in winter (less daylight)
🌙 **Moon**	May rise and set at different times depending on phase	Goes through all 8 phases in ~29.5 days (new → crescent → quarter → gibbous → full → gibbous → quarter → crescent → new)	Cycle repeats ~12–13 times per year

⭐ **Stars**	Not visible in daytime (outshone by sun). Appear at dusk, move slowly through night.	Constellations shift position as Earth orbits, but patterns stay the same	Different constellations visible in different seasons (Earth's orbital position changes which stars face our night side)

📦 Materials

🏠 What You'll Need

- This worksheet (printed)
- Crayons or colored pencils
- Today's moon phase (parent looks up or child knows from Moon Journal)
- Today's season and approximate sunrise/sunset times

🔬 The Pattern Prediction Challenge

Scientists don't just describe what they've seen — they use patterns to *predict* what will happen next. That's what you'll do today! Using what you know about sky patterns, you'll predict what the sky will look like at a specific time.

Step 1 — Gather Your Information

Before predicting, you need data! Fill in:

- Today's moon phase: ________________________
- Today's season: ________________________
- Today's approximate sunrise: _______ Sunset: _______
- Time of year (month): ________________________

Step 2 — Prediction Challenge A: Tonight at 9pm

Based on what you know, predict what the sky will look like tonight at 9pm. Consider:

- Will the sun be visible? (Check sunset time!)
- What phase is the moon in? Will it be visible?
- Will stars be visible? (What season is it?)
- What color will the sky be?

Draw your prediction in the prediction box on the worksheet!

Step 3 — Prediction Challenge B: 6 Months From Now

Think ahead 6 months. What season will it be? Will days be longer or shorter than now? What will the moon look like in 6 months? (Hint: in exactly 6 months the moon may be in approximately the opposite phase from now, or similar — it completes a full cycle every 29.5 days.) Make your prediction!

Step 4 — Check Tonight's Prediction!
Tonight at 9pm, go look at the sky. How did your prediction compare to reality? What did you get right? What surprised you? Fill in the "What I Actually Saw" section on the worksheet.

💡 **This is real science!** Astronomers use exactly this process — they observe patterns, build models, and make predictions. When predictions match reality, that's strong evidence the model is correct. When they don't match, scientists learn something new. Either way, science moves forward!

💬 Discussion Questions

🗣️ **Talk about these together:**

- Which pattern from this unit did you find most surprising? Why?
- If a friend asked "Why does the moon change shape?", what would you tell them now?
- Can you think of any way sky patterns affect your daily life? (Sleep time, when to play outside, farming, navigation...)
- Ancient people figured out all these patterns by careful observation — without telescopes or computers. What do you think that took?

🚀 What's Next in Earth Science?

You've mastered *patterns in the sky* — one of the foundational ideas of Earth Science. Here are some directions to explore next:

- 🌍 **Earth's Place in Space:** How does Earth compare to other planets in our solar system?
- 🌦️ **Weather Patterns:** How does the sun's energy drive weather on Earth?
- 🌊 **Earth's Surface:** How do land and water shapes affect our environment?
- 🔭 **Telescopes & Tools:** How have tools like telescopes expanded our ability to observe the sky?
- 🚀 **Space Exploration:** What have astronauts and space probes discovered by going beyond Earth?

👩‍🏫 Parent/Teacher Notes

- **This is a capstone lesson:** The goal is synthesis and application, not new content. Help the child connect concepts across all 6 lessons. Ask "what does this remind you of from earlier?"
- **The prediction activity is the heart of this lesson:** Making a prediction and then checking it (Step 4 — going outside at 9pm to verify) is genuine scientific practice. Celebrate this even if the prediction isn't perfect.
- **The "6 months from now" prediction:** This is more abstract. Support the child by asking: "What season will it be in 6 months? Will days be longer or shorter? What do we know about those seasons?" Let them reason through it rather than telling them the answer.
- **Celebrate what they've learned:** This is the end of the unit! Review the Sky Patterns Summary chart together. Ask the child to explain each row to you in their own words — that's the best assessment.
- **Unit assessment idea:** Ask the child to draw the sky at three different times (morning, midday, night) and explain one pattern for each. Can they name the phase of the moon right now and predict what it will look like next week?

🎉 Unit 1 Wrap-Up — Sun, Moon & Stars!

You just completed **6 lessons of Earth Science!** Let's review everything you've learned:

☀️

Lesson 1 — The Sun

Earth spins → sun appears to move. Shadows track the sun's position. Never look directly at the sun!

🌙

Lesson 2 — The Moon

Moon orbits Earth in 29.5 days. Phases show how much lit side we see. Moon reflects sunlight.

⭐

Lesson 3 — Stars

Stars are always there — the sun outshines them during the day. Every star is a distant sun.

📅

Lesson 4 — Daylight

Summer = more daylight, winter = less. Earth's tilt (not distance!) causes seasons.

🌅

Lesson 5 — Sky Changes

Sky color follows a daily pattern. Blue at midday, orange/red at sunrise and sunset (Rayleigh scattering).

🔭

Lesson 6 — Patterns

Scientists use patterns to make predictions! Sky patterns are reliable, repeating, and predictable.

🌟 Big Idea of the Unit 🌟

The sun, moon, and stars follow *predictable patterns* that scientists can observe, describe, and use to make accurate predictions. Understanding these patterns is one of humanity's oldest and most important scientific achievements.

🔭 Student Worksheet — Pattern Prediction Challenge

Unit 1, Lesson 6 — Patterns in the Sky

Name: ________________________________ Date: ________________

📋 My Sky Data (fill in before predicting!)

Today's Moon Phase: ______________	**Today's Season:** ______________
Today's Sunrise: ______________	**Today's Sunset:** ______________

🌙 Prediction A: What will the sky look like tonight at 9pm?

My Prediction:

Draw the sky at 9pm here

Color, moon phase, stars, anything else?

Will the sun be visible at 9pm? ____

Will the moon be visible? ____

Will stars be visible? ____

Sky color: ____

What I Actually Saw at 9pm:

Draw what you really saw here

Go outside at 9pm tonight to check!

Sun visible? ____

Moon visible? ____

Stars visible? ____

Sky color: ____

📅 Prediction B: What will the sky be like in 6 months?

Think about: What season will it be? Will there be more or less daylight? What moon phase might we see?

In 6 months it will be: Season: ______________ Month: ______________

Will there be more or less daylight than today? (circle): **MORE** / **LESS** / **ABOUT THE SAME**

Why? ____

Draw what you predict the sky will look like at 9pm in 6 months

✅ Unit Review — What Do You Know?

Match each object with one true fact about it! Draw a line to connect them.

Object	Fact
☀️	Reflects sunlight. Goes through phases in about 29.5 days.
🌙	Always shining! Just too faint to see during the day because of the sun.
⭐	Appears to move east to west each day because Earth is rotating.

✂️ PARENT ANSWER KEY — Detach before giving worksheet to students

🌙 Prediction A — Tonight at 9pm

Answers will vary based on actual conditions. Guide the child to think through:

- **Sun:** Not visible at 9pm (sun sets well before 9pm in most seasons)
- **Moon:** Depends on current phase. If full moon → bright and visible. If new moon → not visible. If quarter → visible part of the night.
- **Stars:** Visible (if sky is clear and 9pm is fully dark — check your location's dusk time)
- **Sky color:** Dark blue to black

📅 Prediction B — 6 Months From Now

Answers depend on current month:

- If currently winter (Dec–Feb) → in 6 months it's summer → *more* daylight
- If currently summer (Jun–Aug) → in 6 months it's winter → *less* daylight
- If currently spring or fall → in 6 months it's the opposite equinox → *about the same* daylight

✅ Unit Review Matching Answers

- ☀️ Sun → "Appears to move east to west each day because Earth is rotating."
- 🌙 Moon → "Reflects sunlight. Goes through phases in about 29.5 days."
- ⭐ Stars → "Always shining! Just too faint to see during the day because of the sun."

🏆 Unit 1 — Full Success Criteria

A child who has mastered this unit can:

- Explain why the sun appears to move from east to west (Earth's rotation)
- Name at least 4 moon phases in correct order
- Explain why stars are invisible during the day
- State that summer has more daylight than winter
- Describe at least 2 different sky appearances throughout a day
- Make a prediction about the sky using known patterns
- Apply the idea that patterns allow predictions — the heart of scientific thinking

🌟 Congratulations on completing Unit 1!

Your child has explored the sky as a real scientist — making observations, identifying patterns, and using those patterns to make predictions. These skills are the foundation of all scientific thinking and will serve them across every subject they study.

🧪 Ready to test your knowledge?

Try the Unit 1 quiz — 10 questions covering everything from the sun's daily motion to moon phases and seasons!

☀️🌙⭐

🌙 Patterns in the Sky

SUN • MOON • STARS • SEASONS

☀️ THE SUN

- Nearest star to Earth
- Rises east, sets west
- Highest at noon
- ⚠️ NEVER look directly at sun

🌙 THE MOON

- Orbits Earth
- Reflects sunlight
- Phases change shape
- 🌑 New → 🌓 Half → 🌕 Full → back

⭐

🍂

⚠️ NO SUN	⚠️ NO MOON	⚠️ NO STAR PATTERNS	⚠️ NO SEASONS
Earth freezes, life dies	Different ocean tides	Couldn't navigate at sea	Same temperature always

- ☀️ Light takes 8 minutes to reach Earth
- 🌙 Same side of Moon always faces us
- 🌟 You see stars from thousands of years ago
- 🌍 Earth orbits the sun in exactly 365.25 days

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☀️🌙⭐ Sun, Moon & Stars Quiz

Grade 1 · Earth & Space Science Unit 1

☀️ **Remember: NEVER look directly at the sun! It can hurt your eyes.**

1 ☀️ **Why does the sun seem to move across the sky — rising in the morning and setting at night?**

- The sun travels around the Earth every day.
- The Earth spins around, so the sun looks like it moves.
- Clouds push the sun across the sky.
- The moon pulls the sun from one side to the other.

2 👁️ **What should you NEVER do when studying the sun?**

- Draw a picture of the sun.
- Look directly at the sun with your eyes.
- Use a shadow stick to track the sun.
- Read books about the sun.

3 🌙 Why does the moon seem to change shape each night?

- The moon is slowly shrinking and growing.
- Clouds cover different parts of the moon.
- We see different amounts of the moon's lit side as it moves around Earth.
- The moon turns off its light sometimes.

4 ⭐ Why can't we see stars during the daytime?

- Stars fly away when the sun comes up.
- Stars only exist at night.
- The bright sunlight makes stars too faint to see, but they're still there.
- Stars go to sleep during the day.

5 🌤️ Why is the daytime sky blue?

- The ocean reflects blue light up into the sky.
- Air in the sky scatters blue light from the sun in all directions.
- The sky is painted blue by clouds.
- Blue is the color of space.

6 🌅 Why does the sky look orange, pink, or red at sunrise and sunset?

- The sun changes color in the morning and evening.
- Sunlight has to travel through more air near the horizon, scattering away blue light so reds and oranges remain.
- Pollution from factories colors the sky orange.
- The moon reflects orange light at those times.

7 ☀️❄️ In which season does your part of Earth get the MOST hours of daylight?

- Winter
- Spring
- Summer
- The same every season

8 🌍 What causes the seasons to change — making summer hot and winter cold?

- Earth moves closer to the sun in summer and farther away in winter.
- Earth is tilted, so different parts get more direct sunlight at different times of year.
- The sun gets bigger in summer and smaller in winter.
- Clouds block the sun more in winter.

9 🌑 When we see a New Moon, what is really happening?

- The moon has flown out of the sky.
- Earth's shadow is covering the moon.
- The moon is between Earth and the sun, so its dark side faces us.
- The moon exploded and is growing back.

10 🌟 Which of these is a pattern we can predict about the sky?

- Tomorrow the sky might randomly turn green.
- The moon will go from full to new and back again in about 29.5 days.
- Stars move to different constellations every week.
- The sun rises in the west every morning.

Sun, Moon & Stars Unit Quiz Answer Key

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1. undefined

Answer: B. The Earth spins around, so the sun looks like it moves.

🌍 The Earth spins like a top! As our planet rotates, the sun appears to move from east to west across the sky. The sun doesn't actually move — WE do!

2. undefined

Answer: B. Look directly at the sun with your eyes.

⚠️ NEVER look directly at the sun! The sun's light is so powerful it can damage your eyes — even on a cloudy day. Always study the sun safely, like using shadows.

3. undefined

Answer: C. We see different amounts of the moon's lit side as it moves around Earth.

🌕 The moon doesn't make its own light — it reflects sunlight! As the moon travels around Earth, we see different portions of its lit side. That's what creates the phases: new moon, crescent, half, gibbous, and full!

4. undefined

Answer: C. The bright sunlight makes stars too faint to see, but they're still there.

✨ Stars are always shining! But the sun is so bright that its light floods the sky during the day, making the faint light from faraway stars impossible to see. At night, with the sun on the other side of Earth, we can finally spot them!

5. undefined

Answer: B. Air in the sky scatters blue light from the sun in all directions.

💙 Sunlight contains all the colors of the rainbow! When sunlight hits air, tiny particles scatter blue light in every direction much more than other colors. This is called Rayleigh scattering, and it's why the whole daytime sky looks blue!

6. undefined

Answer: B. Sunlight has to travel through more air near the horizon, scattering away blue light so reds and oranges remain.

🌄 Near the horizon, sunlight travels a longer path through Earth's atmosphere. By the time it reaches your eyes, most of the blue light has scattered away — leaving the warmer reds, oranges, and pinks. Beautiful science!

7. undefined

Answer: C. Summer

🌞 Summer! During summer, your part of Earth is tilted toward the sun, so the sun takes a longer path across the sky. That means more hours of daylight — and longer, brighter days!

8. undefined

Answer: B. Earth is tilted, so different parts get more direct sunlight at different times of year.

🌎 Seasons are caused by Earth's TILT — not its distance from the sun! Earth is tilted at about 23.5 degrees. When your part of Earth is tilted TOWARD the sun, sunlight hits more directly → summer. Tilted AWAY → winter. (Fun fact: Earth is actually slightly closer to the sun in January — winter in the Northern Hemisphere!)

9. undefined

Answer: C. The moon is between Earth and the sun, so its dark side faces us.

🌑 During a New Moon, the moon is positioned between Earth and the sun. The side of the moon facing us is in shadow, so we can't see it at all! The moon is still there — we just see its dark side. About 29.5 days later, it will be a Full Moon again!

10. undefined

Answer: B. The moon will go from full to new and back again in about 29.5 days.

📅 Patterns let us make predictions! The moon follows a reliable 29.5-day cycle of phases. Scientists — and YOU — can use this pattern to predict what the moon will look like any night. The sun also rises in the EAST every day, another reliable sky pattern!